NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

DEPARTMENT OF
NAME OF THE DEPARTMENT

A VERY LONG AND IMPRESSIVE THESIS TITLE WITH A FORCED LINE BREAK

JOHN VERY LONGNAME DOE
Master/BSc in Name of Previous Degree

DOCTORATE IN STUDY PROGRAM NAME
SPECIALIZATION IN SPECIALIZATION NAME

NOVA University Lisbon
January, 2000

NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

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January, 2000

A Very Long and Impressive Thesis Title with a Forced Line Break

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This document was produced with the luaLaTeX [1] typesetting system and the NOVAthesis template (v7.10.7) [2].

[1] The LaTeX Project team. *LaTeX – A document preparation system*. 1985. URL: `https://www.latex-project.org`.

[2] J. M. Lourenço. *The NOVAthesis LaTeX Template User's Manual*. NOVA University Lisbon. 2021. URL: `https://github.com/joaomlourenco/novathesis/raw/main/template.pdf`.

Dedicatory lorem ipsum.

Acknowledgements

Acknowledgments are personal text and should be a free expression of the author.

However, without any intention of conditioning the form or content of this text, I would like to add that it usually starts with academic thanks (instructors, etc.); then institutional thanks (Research Center, Department, Faculty, University, FCT / MEC scholarships, etc.) and, finally, the personal ones (friends, family, etc.).

But I insist that there are no fixed rules for this text, and it must, above all, express what the author feels.

"You cannot teach a man anything; you can only help him discover it in himself."

— **Galileo**, Somewhere in a book or speach
(Astronomer, physicist and engineer)

Abstract

Regardless of the language in which the dissertation is written, usually there are at least two abstracts: one abstract in the same language as the main text, and another abstract in some other language.

The abstracts' order varies with the school. If your school has specific regulations concerning the abstracts' order, the NOVAthesis LaTeX (novathesis) (LaTeX) template will respect them. Otherwise, the default rule in the novathesis template is to have in first place the abstract in *the same language as main text*, and then the abstract in *the other language*. For example, if the dissertation is written in Portuguese, the abstracts' order will be first Portuguese and then English, followed by the main text in Portuguese. If the dissertation is written in English, the abstracts' order will be first English and then Portuguese, followed by the main text in English. However, this order can be customized by adding one of the following to the file `5_packages.tex`.

```
\ntsetup{abstractorder={<LANG_1>,...,<LANG_N>}}
\ntsetup{abstractorder={<MAIN_LANG>={<LANG_1>,...,<LANG_N>}}}
```

For example, for a main document written in German with abstracts written in German, English and Italian (by this order) use:

```
\ntsetup{abstractorder={de={de,en,it}}}
```

Concerning its contents, the abstracts should not exceed one page and may answer the following questions (it is essential to adapt to the usual practices of your scientific area):

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution/contribution?
4. What results (implications/consequences) from the solution?

Keywords: One keyword · Another keyword · Yet another keyword · One keyword more · The last keyword

Sustainable Development Goals (SDG):

Resumo

Independentemente da língua em que a dissertação está escrita, geralmente esta contém pelo menos dois resumos: um resumo na mesma língua do texto principal e outro resumo numa outra língua.

A ordem dos resumos varia de acordo com a escola. Se a sua escola tiver regulamentos específicos sobre a ordem dos resumos, o template (LaTeX) novathesis irá respeitá-los. Caso contrário, a regra padrão no template novathesis é ter em primeiro lugar o resumo *no mesmo idioma do texto principal* e depois o resumo *no outro idioma*. Por exemplo, se a dissertação for escrita em português, a ordem dos resumos será primeiro o português e depois o inglês, seguido do texto principal em português. Se a dissertação for escrita em inglês, a ordem dos resumos será primeiro em inglês e depois em português, seguida do texto principal em inglês. No entanto, esse pedido pode ser personalizado adicionando um dos seguintes ao arquivo `5_packages.tex`.

```
\abstractorder(<MAIN_LANG>):={<LANG_1>,...,<LANG_N>}
```

Por exemplo, para um documento escrito em Alemão com resumos em Alemão, Inglês e Italiano (por esta ordem), pode usar-se:

```
\ntsetup{abstractorder={de={de,en,it}}}
```

Relativamente ao seu conteúdo, os resumos não devem ultrapassar uma página e frequentemente tentam responder às seguintes questões (é imprescindível a adaptação às práticas habituais da sua área científica):

1. Qual é o problema?
2. Porque é que é um problema interessante/desafiante?
3. Qual é a proposta de abordagem/solução?
4. Quais são as consequências/resultados da solução proposta?

Palavras-chave: Primeira palavra-chave · Outra palavra-chave · Mais uma palavra-chave · A última palavra-chave

Objetivos de Desenvolvimento Sustentável (ODS):

Disclosure of Usage of Generative Artificial Intelligence

The author, John Very Longname Doe, acknowledges the use of the following generative artificial intelligence tools in the development of this thesis. These technologies were utilized under strict supervision, and all AI-generated outputs were critically reviewed and verified for accuracy. The author accepts full responsibility for the validity and originality of the content, confirming that the use of these tools aligns with all institutional policies and standards of academic integrity.[1]

Generative AI tools used: ChatGPT-5.2, Gemini 3 Pro, Claude 4.5 Opus, and Perplexity 12.12.25.

Activities supported by Generative Artificial Intelligence Tools:[2]

Conceptualization

- ☐ Idea generation
- ☑ **Defining the research objective**
- ☐ Formulating research questions and hypotheses
- ☑ **Feasibility assessment and risk evaluation**
- ☐ Hypothesis viability assessment
- ☐ Simulated debate and argument testing

Literature Review

- ☐ Search and Discovery
- ☐ Literature summarization and synthesis
- ☑ **Concept Mapping and Systematization**
- ☐ Market/patent landscape analysis
- ☐ Gap Identification and Novelty Evaluation
- ☐ Cross-lingual literature comprehension

Methodology

- ☐ Experimental Design Optimization
- ☐ Development of experimental or research protocols
- ☑ **Methodological Instrument Selection**

[1] This declaration was generated using LuaLaTeX [15] and the LaTeX package `aidisclose` [8].
[2] This list extends the GAIDeT taxonomy [14].

Software Development and Automation

- [ ] Code Generation
- [x] **Debugging and Repair**
- [ ] Algorithm design
- [ ] Refactoring and Optimization
- [x] **Process automation**
- [ ] Code documentation and comment generation

Data Management

- [ ] Data collection
- [ ] Data analysis
- [ ] Synthetic data generation
- [ ] Validation
- [x] **Quantitative Plotting and Charting**
- [ ] De-identification and anonymization support
- [ ] Data cleaning
- [ ] Reproducibility and rerun checks
- [x] **Audio-to-text transcription and diarization**
- [x] **Data curation and organization**
- [ ] Data labeling and annotation assistance

Visuals and Multimedia

- [ ] Synthetic Asset Generation
- [x] **Image Enhancement and Editing**
- [ ] Diagrammatic and Schematic Design

Writing and Editing

- [ ] Drafting Text
- [ ] Formulation of conclusions
- [ ] Citation formatting and bibliography management
- [ ] Polishing and Editing
- [ ] Tone Adjustment
- [x] **Press releases and outreach materials**
- [ ] Abstract and Executive Summary Generation
- [ ] Translation
- [ ] Title Generation

Ethics Review

- [x] **Bias analysis and discrimination assessment**
- [ ] Ethical risk analysis
- [ ] Data confidentiality monitoring
- [ ] Monitoring compliance with ethical standards

Quality Assurance

- [x] **Simulated Peer Review**
- [ ] Identification of limitations
- [ ] Publication support
- [ ] Consistency checking against field trends
- [x] **Publication strategy and journal selection**

Additional comment #1: The AI was used primarily for refining the code in Section 3, but the algorithm logic was derived manually.

Additional comment #2: This is a second additional comment!

Additional comment: This is a third additional comment! It is starred and, thus, does not get a number. Also, this comment is longer and, hopefully, will span to the next line!

Additional comment #3: This is the third non-starred additional comment!

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Acronyms

DI	Department of Computer Science *(p. 1)*
FCT	NOVA School of Science and Technology *(pp. 1, 7)*
NOVA	NOVA University Lisbon *(p. 1)*
novathesis	NOVAthesis LaTeX *(pp. vi, viii, xvi, 1–5)*

1

Introduction

This is the novathesis LaTeX template Version 7.10.7 from Template!date2026-03-25.

This work is licensed under the LaTeX Project Public License v1.3c. To view a copy of this license, visit the LaTeX project public license.

This first Chapter introduces the novathesis template and how it is organized. In Chapter 2 you can find some specific instructions on how to use this template. Chapter 4 shows some examples and give some hints on how to write your text. Please read these next Chapters carefully.

1.1 The *NOVAthesis* Template

The novathesis template was born at the Department of Computer Science (DI) of NOVA School of Science and Technology (FCT) of NOVA University Lisbon (NOVA), Portugal. But the user base grew… initially grew to other Departments of NOVA FCT, then to other Schools of NOVA, and later to other Schools of other Universities. Currently more than 25 Schools are natively supported by the novathesis template. The full list of supported Schools can be founf at the Project's GitHub page.

1.1.1 Your Time is Precious

Did you learn how to drive by sitting by the wheel and throwing your car into the road? Most probably you did take your time *learning the rules* and *practicing* first! Likewise, it is not wise to throw yourself at the task of writing a thesis/dissertation in LaTeX without seriously considering the following recommendation!

> If you are going to spend zillions of hours writing your thesis/dissertation using the novathesis LaTeX template (or some other LaTeX template), be wise and spend a couple of hours learning how to use it properly by reading this manual. And then, be even wiser, and spend a few more hours learning some LaTeX. I am sure that the time you are investing now will pay itself countless times before you submit your thesis/dissertation.
>
> — *João Lourenço*

1.1.2 Recognition

The novathesis template was born in 1996, and what you see now accumulates to many many hundreds (thousands?!) of working hours, unpaid and stolen from family and friends. This work is available to the community under the LaTeX Project Public License v1.3c, which means you are entitled to use it for free and change it at your will. However, if you decide to use this template to write your thesis/dissertation, **be fair to the developers** and:

1. Cite the novathesis manual [9] in a place of your choice (e.g., in the *Acknowledgments*) of your thesis/dissertation with "`\cite{novathesis-manual}`" . If you cite it this way, the correct entry will be added automatically to your bibliography (no need to worry with the necessary BibTeX entry, as it will be added automatically);

2. Go to the project web page in GitHub and give the project a star (marked with a red ellipse at the top-right in Figure 1.1); and

3. Make a donation by visiting the novathesis project page and clicking in the button marked with a green ellipse at the top-center in Figure 1.1). Alternatively, just click **HERE** and your browser will be directed to the right page.

1.2 Getting Started

The template provides an *easy to use* setting for you to write your thesis/dissertation in LaTeX:

- Select your school;
- Fill your thesis metadata (title, research field, etc) in the file "`template.tex`";
- Create your thesis/dissertation contents using the files in folder "`Chapters`"; and
- Process using you favorite LaTeX processor (pdfLaTeX, or XeLaTeX).

1.2.1 Using Overleaf

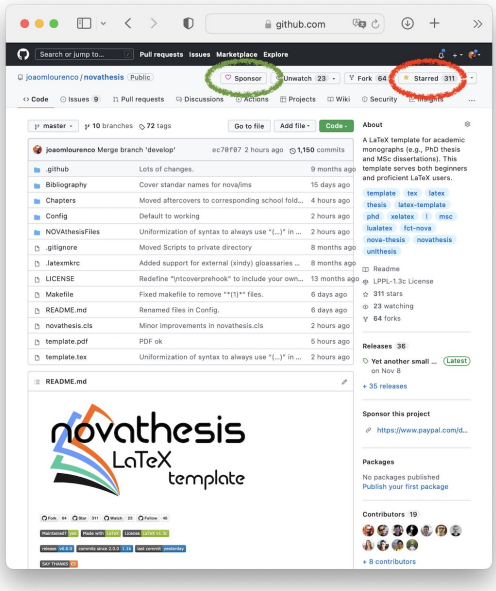


Figure 1.1: The novathesis project web page in GitHub.

Overleaf is a collaborative cloud-based LaTeX editor used for writing, editing and publishing scientific documents. Like "Google Docs", for LaTeX users. You can edit and compile your LaTeX source on the cloud, without installing software in your own computer, and, much like *Google Docs*, you can share your document with others users and everybody can edit the same file at the same time (this may be dangerous).

If you do not have an account in Overleaf, you must create one first.

Figure 1.2: NOVAthesis template in Overleaf.

Once you have an account, please access the novathesis template in Overleaf and select the green button *Open as Template* (see Figure 1.2).

Please notice that the version currently available in Overleaf (v7.1.18) is slightly outdated (current version is v7.10.7). Follow the steps below to use the latest version (highly recommended):

1. *Download the latest version from the GitHub repository as a Zip file.*
2. *Login to your favorite LaTeX cloud service. I recommend Overleaf but there are alternatives (these instructions apply to Overleaf and you'll have to adapt for other providers).*
3. *In the menu select:* `New project` → `Upload project`.
4. *Upload the zip file.*
5. *Select "template.tex" as the main file.*
6. *Let Overleaf compile the document.*

Notice that you need a (student) subscription to compile the novathesis template in Overleaf, otherwise your compilation will always time out.

1.2.2 Using a Local LaTeX Installation Local

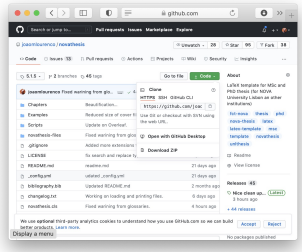


Figure 1.3: The NOVAthesis Project page in GitHub.

First of all, start by installing LaTeX in your computer. There are two main distributions, MikTeX and TeXLive, and both of them are available for the 3 most popular Operating Systems: Linux, macOS and Windows.

Be aware that a full installation of MikTeX or TeXLive will take near 5 GB of hard disk space. So, think twice before installing the full distribution. See the novathesis Wiki for the list of packages required to compile the template.

Once you have LaTeX up and running, remember to install a good LaTeX text editor. I recommend you to take a look at this post in the `tex.stackexchange.com` site. If you want a quick and dirty recommendation, try TeXStudio.

Now, you must access the novathesis repository in GitHub, select the green button *Code* and then *download* (or *clone*) the template. You will always get the latest version of the template (currently v7.10.7 from 2026-03-25).

1.3 Getting Help

No! You don't have to use this template to write your thesis. You don't even have to use LaTeX. However, writing a thesis is serious stuff, and which tool you shall use to write it is not a decision to make lighthearted.

LaTeX is hard enough by itself. This template aims at making your life easier, but not easy. If you choose to use this template to write your thesis, you are very welcome. However, don't expect me to provide you help with LaTeX. Look for help with your friends (you have some friends, don't you?), or search the web, or try even to read some book(s) on LaTeX. In the end you will certainly find the experience rewarding.

When you come to the point of "*How do I do this with the* novathesis *template?*", remember…

1. To check the novathesis wiki and have some hope! `:D`

2. Google is your best friend.

3. Search the GitHub Discussions page for a question related to yours. *If and only if* you don't find one, then post your own question in English please!

4. Search the NOVAtheis Facebook group for a question related to yours. *If and only if* you don't find one, then post your own question in either Portuguese or English, at your preference.

When you post your own question, remember to **always** state the novathesis version number you are using and referring to.

Please do not attempt to contact me directly (email, Messenger, etc)…
I WILL NOT REPLY!

1.3.1 Suggestions, Bugs and Feature Requests

Help: If you just need some help, see above Section 1.3.

Suggestion: Do you have a suggestion/recommendation? Please add it to the wiki and help other users!

Bug: Did you find a bug? Please open an issue. Thanks!

New Feature: Would you like to request a new feature (or support of a new School)? Please open an issue. Thanks!

1.4 Donors

The list of *Donnors* is available in the novathesis Project page.

1.5 Disclaimer

Although the novathesis template is endorsed by some Schools (e.g., linked from NOVA FCT web site), the novathesis template **this not an official template** for any School.

The novathesis template exists to make your life easier and we do our best to make it compliant to the supported (+25) Schools' regulations but, in the end of the line, you and only you are accountable for both the look and the contents of the document you submit as your thesis/dissertation.

2

NOVAthesis Template *User's Manual*

2.1 Introduction

This chapter is a complete guide to the novathesis template. It covers everything from the first-time setup to the fine-tuning of covers, fonts, citation styles, and the document workflow. All user customization flows through the files inside the `0-Config/` folder — the rest of the project tree is never edited directly.

For a reference to the automated build assistant `.Build/build.py`, see Chapter A. For a guide to the cover page system, see Chapter **??**.

2.2 Getting Started

2.2.1 Local LaTeX Installation

novathesis requires a *full* LaTeX installation. The two major distributions are:

TeXLive The reference distribution maintained by LaTeX developers. The full scheme occupies around 5 GB but includes every package you will ever need. Strongly recommended unless disk space is very tight. Download from `https://www.tug.org/texlive/`.

MikTeX Installs a minimal base and downloads missing packages on the fly. A good choice when disk space is limited. Packages installed on the fly may occasionally fail, so keep the installation up to date. Download from `https://miktex.org/`.

After installation, update all packages via the *TeXLiveManager* or the *MikTeXConsole* before compiling the template for the first time.

Recommended engine. LuaLaTeX is the recommended engine for most schools because it handles Unicode fonts natively. XeLaTeX is a close alternative. pdfLaTeX works but cannot load OpenType/TrueType fonts.

School-specific fonts. Some Schools or font-styles require fonts not available in LaTeX. In normal conditions, they will be downloaded automatically. However, if they are not, you may download the necessary fronts from `https://github.com/joaomlourenco/novathesis-extras/tree/main/Fonts`, unzip (if necessary), and copy the `*.ttf` files to `NOVAthesisFiles/FontStyles/Fonts/`.

2.2.2 Downloading the Template

Clone the GitHub repository (recommended — you get all update history):

```
git clone --depth=1 https://github.com/joaomlourenco/novathesis.git
```

Or download the latest ZIP directly:

```
# Click "Code > Download ZIP" on https://github.com/joaomlourenco/novathesis
```

The top-level folder is your working directory. Rename it to anything you like. Open `template.tex` in your editor and compile it — you should get a PDF immediately with the default settings (NOVA School of Science and Technology (FCT) PhD in English).

2.2.3 Overleaf

Overleaf users can import the template directly from the GitHub link. Once imported:

1. Set the *Compiler* to **LuaLaTeX** (Menu → Compiler).
2. Set the *Main document* to `template.tex`.
3. Set the *Bibliography tool* to **Biber** (Menu → Bibliography).
4. Press *Recompile*.

On Overleaf you cannot run `make` or `build.py`; all customization must be done by editing the `0-Config/` files directly (see Section 2.3).

2.3 Project Structure

Table 2.1 summarises the folders and top-level files in the project. Items marked **(edit)** are intended for user customization; items marked **(do not edit)** are part of the template engine.

2.3.1 The `0-Config/` Files

All customization lives in `0-Config/`. Table 2.2 describes each file's purpose.

The *active* setting for any option is the *uncommented* line; commented-out lines show available values as documentation:

```
% \ntsetup{doctype=msc}     % commented = not active
\ntsetup{school=nova/fct}  % uncommented = active
```

Table 2.1: Project structure and access guidelines.

Path	Purpose	Access
`template.tex`	Main document entry point	do not edit
`novathesis.cls`	Document class	do not edit
`0-Config/`	All user configuration	**edit**
`1-FrontMatter/`	Abstracts, glossaries, front pages	**edit**
`2-MainMatter/`	Thesis chapters	**edit**
`3-BackMatter/`	Appendices and annexes	**edit**
`4-Bibliography/`	`.bib` database files	**edit**
`5-Figures/`	Images	**edit**
`NOVAthesisFiles/`	Template internals (styles, schools, strings	do not edit
`Scripts/`	Utility scripts	do not edit
`.Build/`	Build assistant (`build.py`)	do not edit

Table 2.2: Configuration files in `0-Config/`.

File	Purpose
`1_novathesis.tex`	Document options (`\ntsetup{...}`)
`2_biblatex.tex`	Bibliography/biblatex configuration
`3_cover.tex`	Cover metadata (title, author, advisers, committee)
`4_files.tex`	List of content files to include
`5_packages.tex`	User-added packages and custom macros
`9_*.tex`	School-specific overrides (department, degree names)

2.4 Configuring the Document (`1_novathesis.tex`)

2.4.1 Essential Options

`doctype` The type of document being written. Default: `phd`.

```
\ntsetup{doctype=phd}       % PhD thesis
\ntsetup{doctype=phdplan}   % PhD thesis plan
\ntsetup{doctype=phdprop}   % PhD thesis proposal
\ntsetup{doctype=msc}       % MSc dissertation
\ntsetup{doctype=mscplan}   % MSc dissertation plan
\ntsetup{doctype=bsc}       % BSc report
\ntsetup{doctype=plain}     % Other report
```

The `*plan` and `*prop` document types produce a *simplified* front matter: only the abstracts and the AI disclosure statement are included. Pages such as the dedication, acknowledgements, quote, and evaluation committee are suppressed regardless of the `docstatus` setting.

`school` The school identifier. This selects the institutionally-compliant cover layout. Default: `nova/fct`.

```
\ntsetup{school=nova/fct}       % NOVA FCT (default)
\ntsetup{school=nova/fcsh}      % NOVA FCSH
\ntsetup{school=nova/itqb/green}
\ntsetup{school=ulisboa/ist}
\ntsetup{school=uminho/eeng}
% ... see 1_novathesis.tex for the full list
```

docstatus The workflow stage. Controls which front-matter pages are included. Default: `working`. See Section 2.9 for the full implications of each value.

```
\ntsetup{docstatus=working}       % drafting (default)
\ntsetup{docstatus=provisional}   % submitted; no committee yet
\ntsetup{docstatus=final}         % post-examination; committee, book spine,
↪  PDF/A-2b
```

lang Main language of the document body (ISO 3166-1 alpha-2 code). Default: `en`.

```
\ntsetup{lang=en}    % English (default)
\ntsetup{lang=pt}    % Portuguese
\ntsetup{lang=de}    % German
% cat, cz, dk, es, fr, gr, it, nl, pl, sk, uk also available
```

media Target output medium. Default: `screen`.

```
\ntsetup{media=screen}  % equal margins, colored links (default)
\ntsetup{media=paper}   % binding margins, black links
```

2.4.2 Appearance Options

style/font Body font family. Default: `newpx` (Palatino-like). Options marked with (*) require LuaLaTeX or XeLaTeX.

```
\ntsetup{style/font=newpx}         % Palatino-like (default)
\ntsetup{style/font=libertine}     % Linux Libertine
\ntsetup{style/font=opensans}      %
\ntsetup{style/font=arial}         % (*) Arial
\ntsetup{style/font=calibri}       % (*) Calibri
\ntsetup{style/font=newsgott}      % (*) NewsGotT (UMINHO)
% Any standard LaTeX font package name is also accepted.
```

For system fonts (LuaLaTeX/XeLaTeX only), provide the font name and attributes explicitly:

```
\ntsetup{style/font=Times New Roman}
\ntsetup{style/font/attributes={
  Extension       = .ttf,
  Ligatures       = TeX,
  UprightFont     = *,
  ItalicFont      = * Italic,
```

```
  BoldFont        = * Bold,
  BoldItalicFont  = * Bold Italic,
}}
```

`style/chapter` Chapter heading style. Default: `bar`.

```
\ntsetup{style/chapter=bar}          % (default)
\ntsetup{style/chapter=bar-compact}  % less vertical space
\ntsetup{style/chapter=elegant}      % classical
\ntsetup{style/chapter=bluebox}
\ntsetup{style/chapter=compact}      % minimal vertical space
\ntsetup{style/chapter=ist}          % ULISBOA-IST
\ntsetup{style/chapter=fmv}          % ULISBOA-FMV
```

`color/links` Color of hyperlinks in `screen` media. Default: a dark blue. Any color name from the `xcolor` package works.

```
\ntsetup{color/links=SteelBlue}
\ntsetup{color/links=black}   % effectively disables colored links
```

`style/url` URL font style.

```
\ntsetup{style/url=same}  % use main body font for URLs
```

`numbering/floats` Float (figure/table) numbering scheme. Default: `chapter` (*chapter.number*).

```
\ntsetup{numbering/floats=chapter}     % 1.1, 1.2, 2.1 … (default)
\ntsetup{numbering/floats=sequential}  % 1, 2, 3 …
```

2.4.3 Print Control Options

These options control which supplementary pages are included in the generated PDF. Many have defaults that vary with `school` and `docstatus`:

`print/frontpage` Print a front page (second cover). Default: `false`.

```
\ntsetup{print/frontpage=true}
```

`print/committee` Print the evaluation committee. Default: `false` for `working`/`provisional`; `true` for `final`.

```
\ntsetup{print/committee=true}   % force on
\ntsetup{print/committee=false}  % force off
```

`print/statement` Print the declaration/integrity statement. Default: `false`.

```
\ntsetup{print/statement=true}
```

`print/copyright` Print the copyright notice. Default: `true`.

```
\ntsetup{print/copyright=false}
```

spine/layout Print the book spine. Default: `false` for `working` and `provisional`; `trim` (printed) for `final`.

```
\ntsetup{spine/layout=false}  % suppress spine even in final
\ntsetup{spine/layout=trim}   % force spine even in provisional
```

print/index Print a word index at the end. Default: `false`.

```
\ntsetup{print/index=true}
```

print/timestamp Print a generation timestamp on the cover. Default: `true` for `working`; `false` otherwise.

```
\ntsetup{print/timestamp=false}
```

print/webography Title for a separate bibliography section that collects only `@online` entries. When empty (default), all references are printed together.

```
\ntsetup{print/webography=Webography}
```

print/aidisclosure Print an AI use disclosure statement. Default: `aidisclose` (uses `1-FrontMatter/7-aidisclose.tex` with the `aidisclose` package).

```
\ntsetup{print/aidisclosure=false}            % suppress
\ntsetup{print/aidisclosure=aidisclose}       % default package
\ntsetup{print/aidisclosure=aidisclosure.tex} % custom file
```

2.4.4 Language Options

abstractorder Ordered list of languages for the abstract sequence. Default: `{lang, en}` when `lang≠en`; otherwise `{en, pt}`.

```
\ntsetup{abstractorder={en,pt}}
\ntsetup{abstractorder={en,pt,uk,gr}}   % add Ukrainian & Greek
```

lang/cover Language for the cover and front page. Default: same as `lang`.

```
\ntsetup{lang/cover=pt}   % Portuguese cover even if body is English
```

lang/copyright Language for the copyright message. Default: same as `lang`.

lang/extra Additional languages loaded in the document (besides the main language and abstract languages).

```
\ntsetup{lang/extra={de,es}}
```

2.4.5 Sustainable Development Goals (SDGs)

NOVAthesis can print the United Nation's SDG icons at the bottom of the abstract page(s). All three options below should be set together:

```
\ntsetup{print/sdgs/list={3,7,13}}     % SDG numbers to print
\ntsetup{print/sdgs/type=inverted}     % inverted | normal | mono
\ntsetup{print/sdgs/size=2cm}          % icon size
\ntsetup{print/sdgs/hspace=5mm}        % horizontal gap between icons
\ntsetup{print/sdgs/vspace=4mm}        % vertical gap below icons
```

The full list of SDG numbers and their titles is reproduced in `0-Config/1_novathesis.tex` as comments for quick reference.

2.4.6 Glossary and Miscellaneous Options

```
% Hyperlink color for glossary entries (default: black)
\ntsetup{glossaries/color/gls=SteelBlue}

% Show back-references in glossary entries (default: true)
\ntsetup{glossaries/list/reverse=false}

% Glossary column widths
\ntlatesetup{glossaries/layout={>{\raggedright}p{0.2\textwidth}X}}

% Include the Table of Contents as an entry in itself (default: false)
\ntsetup{tocintoc=true}

% Book spine width (default: 1cm)
\ntsetup{spine/width=2cm}

% Debug helpers (print grid and bounding boxes on cover/spine)
\ntsetup{debug={cover,spine}}
```

2.5 Cover Metadata (`3_cover.tex`)

2.5.1 Title and Subtitle

`\nttitle` accepts an optional *type* and a *language*:

```
\nttitle(main,en){Full Thesis Title}
\nttitle(main,pt){Título Completo da Tese}
\nttitle(sub,en){Optional Subtitle}
\nttitle(sub,pt){Subtítulo Opcional}
% Variants for specific contexts (if needed):
\nttitle(cover,en){Shorter Title for the Cover}
\nttitle(abstract,en){Title Version for the Abstract}
\nttitle(spine,en){Very Short Title}
```

Use \\ to force line breaks anywhere in a title.

2.5.2 Department, Degree, and Specialization

```
\ntdepartment*(en){Name of the Department}
\ntdepartment*(pt){Nome do Departamento}

\ntdegreename*(en){Computer Science}
\ntdegreename*(pt){Engenharia Informática}

\ntspecialization*(en){Distributed Systems}    % omit if not applicable
\ntspecialization*(pt){Sistemas Distribuídos}
```

NOVA FCT users: open `0-Config/9_nova_fct.tex` instead and uncomment the correct department/degree entries.

2.5.3 Author

```
\ntauthorname(m){John Very Longname Doe}{John Doe}
% (m) = male, (f) = female
% First argument: full name for the cover
% Second argument: short name for other uses
```

For the author's previous degree (printed on some covers):

```
\ntauthordegree(en){Master of Science in Computer Science}
\ntauthordegree(pt){Mestrado em Engenharia Informática}
```

2.5.4 Advisers

```
% Principal adviser
\ntaddperson{adviser}(a,f){Mary Adviser, Full Professor, NOVA University Lisbon}
% Co-advisers
\ntaddperson{adviser}(c,m){John Co-Adviser, Associate Professor, IST}
```

Valid role codes: `a` (adviser), `c/co` (co-adviser), `t` (tutor). Gender codes: `m` (male), `f` (female).

2.5.5 Evaluation Committee

Committee members are only printed when `docstatus=final` (or when forced with `\ntsetup{print/committee=true}`):

```
\ntaddperson{committee}(c,m){Chair Name, Full Professor, NOVA University Lisbon}
\ntaddperson{committee}(r,m){Rapporteur, Associate Professor, University of
↪  Porto}
\ntaddperson{committee}(m,f){Member Name, Assistant Professor, University X}
\ntaddperson{committee}(g,f){Guest Member, Position, Institution}
```

Valid roles: `c` (chair), `r` (rapporteur/arguente), `a` (adviser, if in committee), `co` (co-adviser, if in committee), `m` (member), `g` (guest).

2.5.6 Dates and Sponsors

```
\ntdate(submission){2025-09-30}   % YYYY-MM-DD preferred
\ntdate(exam){2026-01-15}

% Sponsors (logos or text, per language)
\ntsponsors(1,logo,en){\includegraphics[height=1.5cm]{republica_portuguesa_CTES}}
\ntsponsors(2,logo,en){\includegraphics[height=1.5cm]{FCT_MEC_logo_positive}}
```

2.6 Bibliography (`2_biblatex.tex`)

novathesis uses `biblatex` with the `biber` backend.

2.6.1 General Behavioral Options

```
\ntbibsetup{
  backend      = biber,        % recommended; 'bibtex' also works
  giveninits   = true,         % print initials only
  maxbibnames  = 1000,         % list all authors (no 'et al')
  backref      = true,         % show page numbers for each citation
  backrefstyle = three,        % compact back-reference style
  urldate      = iso,          % ISO dates for URL access
  date         = iso,          % ISO dates for publications
}
```

2.6.2 Choosing a Citation Style

Uncomment exactly one style block in `0-Config/2_biblatex.tex`:

Numeric (sorted by first author) The default. Citations like [1]:

```
\ntbibsetup{style=numeric-comp, sortcites=true, sorting=nyt}
```

Numeric (citation order) Sequential citations [1], [2], …:

```
\ntbibsetup{style=numeric-comp, sortcites=true, sorting=none}
```

Alpha Citations like [Lou24]:

```
\ntbibsetup{style=alphabetic, sortcites=true, sorting=nyt}
```

Author-year Citations like (Lourenço, 2024):

```
\ntbibsetup{style=authoryear-comp, sortcites=true, sorting=nyt}
```

Discipline-specific IEEE, APA, Vancouver, Chicago are also available — see the style blocks in `2_biblatex.tex`.

2.6.3 Webography (Separate Online Bibliography)

To collect all `@online` entries in a separate section, set a title:

```
\ntsetup{print/webography=Webography}
```

When this option is active, `@online` references are excluded from the main bibliography list and printed in their own section at the same location.

2.7 Adding Content (`4_files.tex`)

All content files are registered in `0-Config/4_files.tex` using `\ntaddfile`.

2.7.1 Bibliography Files

```
\ntaddfile{bib}{bibliography.bib}
\ntaddfile{bib}{extra.bib}   % multiple .bib files are allowed
```

Place `.bib` files in `4-Bibliography/`.

2.7.2 Front Matter

```
\ntaddfile{dedication}{dedication}          % 1-FrontMatter/dedication.tex
\ntaddfile{acknowledgements}{acknowledgements}
\ntaddfile{quote}{quote}
% Dedication, quote, and acknowledgements are skipped in 'working' mode.

% Integrity statement (school-specific files available in NOVAthesisFiles)
\ntaddfile{statement}[en]{univ-school/statement-en}
```

2.7.3 Abstracts

```
\ntaddfile{abstract}[pt]{abstract-pt}   % 1-FrontMatter/abstract-pt.tex
\ntaddfile{abstract}[en]{abstract-en}
% Other supported language codes: de, es, fr, gr, it, nl, uk, zhs, zht
```

The order in which abstracts appear is controlled by `\ntsetup{abstractorder=...}` (Section 2.4.4).

2.7.4 Glossaries, Acronyms, and Symbols

```
\ntaddfile{glossaries}[glossary]{glossary}   % 1-FrontMatter/glossary.tex
\ntaddfile{glossaries}[acronym]{acronyms}
\ntaddfile{glossaries}[symbols]{symbols}
```

Each file uses standard `glossaries` package commands (`\newglossaryentry`, `\newacronym`, etc.).

2.7.5 Main Chapters

```
\ntaddfile{chapter}{chapter-introduction}   %
↪  2-MainMatter/chapter-introduction.tex
\ntaddfile{chapter}{chapter-methods}
\ntaddfile{chapter}{chapter-results}
\ntaddfile{chapter}{chapter-conclusion}
```

Chapters are included in the order they appear in `4_files.tex`. To restrict a chapter to specific document types, pass an optional list:

```
\ntaddfile{chapter}[mscplan,msc]{chapter-proposal}  % only in MSc docs
```

2.7.6 Appendices and Annexes

```
\ntaddfile{appendix}{appendix-a}   % 2-MainMatter/appendix-a.tex
\ntaddfile{annex}{annex-b}         % 2-MainMatter/annex-b.tex
```

The same optional doctype filter applies to appendices and annexes.

2.7.7 Custom Covers

To override an automatically-generated cover with a custom file:

```
\ntaddfile{cover}[1]{cover-page}   % 1-FrontMatter/cover-page.tex
% cover ids: 1 (main cover), 2 (front page), N (back cover), spine
```

2.8 Custom Packages and Commands (`5_packages.tex`)

Add any extra `\usepackage` commands and custom macros in `0-Config/5_packages.tex`. This file is loaded after the template core so any redefinitions here take precedence.

This file also contains the pre-defined `minted` code environments used throughout the demo chapters:

`latexcode` Syntax-highlighted LaTeX listing with a light beige background.

`bashcode` Shell/Bash listing with the same style.

`pythoncode` Python listing with the same style.

`\latexinline{...}` Inline highlighted code fragments.

Use these environments in your chapters for consistent formatting:

```
\ntsetup{school=nova/fct}
```

```
make lua
```

To add a language-string rename (e.g. change the TOC heading):

```
\ntlangsetup{en/contents=Table of Contents}
\ntlangsetup{pt/contents=Índice}
```

2.9 Document Status Workflow

The `docstatus` option controls which pages are included and how the cover is rendered. Table 2.3 summarises the differences.

Table 2.3: Effect of `docstatus` on the generated document.

Feature	`working`	`provisional`	`final`
Timestamp on cover	yes	no	no
Dedication / quote	no	yes	yes
Acknowledgements	no	yes	yes
Integrity statement	no	yes	yes
Evaluation committee	no	no	yes
Book spine	no	no	yes
PDF/A-2b output	no	no	yes

See Section 2.10.5 for details on the PDF/A archival output.

A typical workflow:

1. Write with `docstatus=working` (fast, no distractions).
2. Switch to `provisional` when submitting to the examination jury — a clean PDF without the committee members.
3. Switch to `final` after the examination — includes the committee, the book spine, and produces a PDF/A-2b archival document (see Section 2.10.5).

2.10 Building the Document

2.10.1 Using `make`

The `Makefile` at the project root wraps `latexmk`:

```
make        # auto-detect engine (usually LuaLaTeX)
make lua    # force LuaLaTeX
make xe     # force XeLaTeX
make pdf    # force pdfLaTeX
make view   # build and open the PDF viewer
make clean  # remove auxiliary files
```

2.10.2 Using `latexmk` Directly

```
latexmk -shell-escape -file-line-error -lualatex template
```

Remember to select **biber** (not bibtex) as the bibliography processor — either in your editor preferences or via `latexmk`'s default settings.

2.10.3 Using `build.py`

The `.Build/build.py` script builds for a specific school without modifying the `0-Config/` files manually. It is primarily intended for CI use and for maintainers generating sample PDFs. For full documentation see Chapter A.

Quick example:

```
python3 .Build/build.py nova/fct --doctype phd --lang en --docstatus final
```

2.10.4 Overleaf

On Overleaf the build is triggered by pressing *Recompile*. Ensure the settings described in Section 2.2.3 are applied (LuaLaTeX compiler, Biber bibliography, `template.tex` as main file). Overleaf compiles on their servers so `make` and `build.py` are not available — all configuration is done by editing the `0-Config/` files.

2.10.5 PDF/A Archival Output

When `\ntsetup{docstatus=final}` is set, novathesis automatically produces a **PDF/A-2b** conformant document — the archival-quality format required by most institutional repositories and digital libraries. No configuration beyond setting `docstatus=final` is needed.

How it works (single-pass). novathesis loads the `pdfx` package during the normal LaTeX preamble, just before `hyperref`. This single-pass approach requires no extra compilation run beyond the usual cross-reference and bibliography passes performed by every LaTeX document.

The XMP metadata packet (title, author, language) is written from the cover-page data defined in `0-Config/3_cover.tex` and embedded into the PDF automatically.

PDF/A standard choice. novathesis targets **PDF/A-2b** (not 1b). PDF/A-1b prohibits transparency, which is incompatible with the TikZ cover-page graphics used by virtually all supported schools. PDF/A-2b lifts this restriction while remaining accepted by all major institutional repositories.

XMP metadata. The thesis title, author name, and language are taken directly from the cover-page data in `0-Config/3_cover.tex` and passed to `pdfx`, which embeds them in the required XMP packet. No additional configuration is needed.

Images and colour spaces. The most common cause of PDF/A validation failures is an embedded image with a device-dependent colour space (plain RGB or CMYK) and no ICC profile. PDF/A-2b requires every included image to carry an embedded ICC profile or use a calibrated colour space. Practical guidance by format:

JPEG / JPG Most cameras and photo-editors embed an sRGB ICC profile automatically. Screenshots and diagrams exported from presentation or diagramming software

often do not. Check the source file and add a profile before including it (*e.g.* via GIMP's *Image* → *Mode* → *Assign Colour Profile*, or with ImageMagick: `magick -profile sRGB.icc input.jpg output.jpg`).

PNG Inkscape embeds an sRGB profile when exporting via *File* → *Export PNG Image*. Images from other sources should be checked; if the profile is missing, add it with the same tools as for JPEG above.

TIFF LaTeX engines cannot include TIFF directly. Convert to PDF or PNG with an embedded ICC profile using an image-editing tool (GIMP, ImageMagick, or Photoshop) before including the result.

PDF (figures) An included PDF figure must itself use device-independent colour spaces throughout. Figures produced by Matplotlib, Inkscape, and TikZ are almost always clean. Scanned or converted PDFs may not be — if VeraPDF reports colour-space errors linked to a specific figure, pre-process it with Ghostscript before re-including:

```
gs -dBATCH -dNOPAUSE -sDEVICE=pdfwrite \
   -sColorConversionStrategy=sRGB \
   -o figure-fixed.pdf figure-original.pdf
```

The safest choice is **vector PDF** (TikZ, Inkscape, Matplotlib): it contains no raster data and is intrinsically colour-space clean.

Opting out. To disable PDF/A output even in `final` mode, uncomment the escape hatch in `0-Config/1_novathesis.tex`:

```
\ntsetup{pdfa=false}
```

Verifying compliance. After the final build, validate with:

- **VeraPDF** (open source, `https://verapdf.org`) — the reference PDF/A validator.
- **Adobe Acrobat Pro** — *Tools* → *Print Production* → *Preflight* → *PDF/A compliance*.
- **PDF/A Live!** (online service) — quick check without local installation.

3

Artificial Intelligence Disclosure Statement

The novathesistempalte supports the rendering of a *Artificial Intelligence Disclosure Statement*.

The rendering is cotrolled by the `\ntsetup{print/aidisclosure=OPTION}` option in file `0-Config/1_novathesis.tex`.

Possible values for `OPTION` are:

`false`	Will omit the AI Disclosure Statement.
`filename.tex`	Will render the given file as the AI Disclosure Statement.
`aidisclose`	Will render the *standard* AI Disclosure Statement with the help of the `aidisclose` package.

3.1 Deactivating the AI Disclosure Statement

In file `0-Config/1_novathesis.tex` add the optoin `\ntsetup{print/aidisclosure=false}`.

3.2 Using Your Own AI Disclosure Statement

In file `0-Config/1_novathesis.tex` add the optoin `\ntsetup{print/aidisclosure=filename.tex}`.

3.3 Using the novathesis Standard AI Disclosure Statement

The novathesis template uses the `aidisclose` package [8], which extends the *GAIDeT (Generative AI Delegation Taxonomy)* [14] to automate Generative AI disclosure statements and checklists.

The `aidisclose` configuration file can be found at `0-Config/7_aidisclose.tex`. Although you can edit/change this file, it is strongly recommended that you use the package's companion web site at `aidisclose.org`, illustrated in Figure 3.1.

Fill the Author name and AI tools used, optionally add comments, leave *Configuration & Rendering* as is, and then select the appropriate checkboxes in *Taxonomy of Delegated Tasks*. Finally, press the geen button at the bottom, and replace the novathesisaidisclose configuration file (`0-Config/7_aidisclose.tex`) with the contents given in the web site.

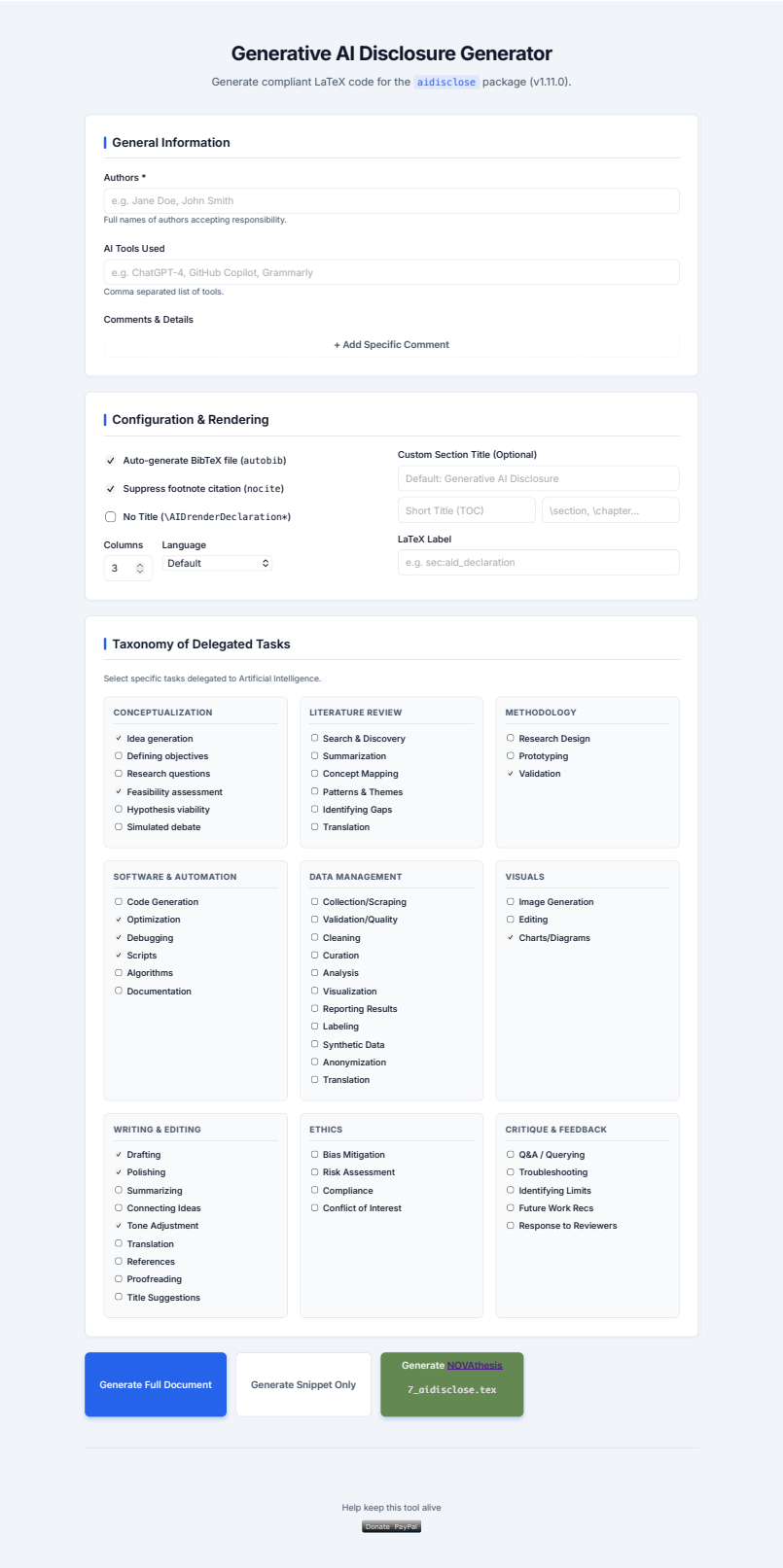


Figure 3.1: The `aidisclose.org` web site

3.3.1 About the AI Tags

3.4 Detailed Taxonomy Descriptions

This appendix provides a detailed breakdown of the `aidisclose` taxonomy. For each category, we define the **Objective** (the high-level goal of using AI in this phase) and the **Scope** (what is generally included or excluded).

1. Conceptualization

Objective: To use AI as a thought partner for brainstorming, refining the research direction, and establishing the theoretical foundation before empirical work begins.

Scope: Includes ideation, hypothesis formation, and feasibility checks. Excludes the actual execution of experiments or data collection.

Keys:

Idea generation: Using AI to brainstorm new research topics, interdisciplinary connections, or novel angles on existing problems.

Objective refinement: Refining vague goals into concrete, actionable research objectives.

Research questions: Drafting and iterating on specific research questions (RQs) to ensure they are clear and answerable.

Feasibility check: Assessing whether the proposed study is viable regarding resources, time, and data availability.

Preliminary research: Conducting quick background checks or "pre-studies" to see if the idea has already been solved.

Simulation/Scenarios: Using AI to conceptualize theoretical models, simulate persona responses, or design hypothetical scenarios.

2. Literature Review

Objective: To accelerate the discovery, synthesis, and organization of existing knowledge.

Scope: Includes searching, summarizing, and translating papers. Excludes the final critical argumentation (which remains the author's responsibility).

Keys:

Search & Discovery: Using AI tools (e.g., semantic search) to find relevant papers that keyword searches might miss.

Summarization: Generating summaries or abstracts of long papers to quickly assess relevance.

Mapping: Visualizing connections, citation networks, or thematic clusters in the literature.

Pattern recognition: Identifying trends or recurring themes across a large corpus of text.

Gap identification: Using AI to suggest areas where current research is lacking or contradictory.

Translation: Translating foreign-language literature to make it accessible for the review.

3. Methodology

Objective: To assist in the structural design of the research study.

Scope: Includes experimental design and instrument creation. Excludes the physical conduct of experiments.

Keys:

Experimental design: Designing the logic, control groups, and variables of the study.

Prototyping: Creating early drafts of survey instruments, interview guides, or experimental apparatus designs.

Method selection: Suggesting appropriate statistical methods or qualitative frameworks for the data.

4. Software Development and Automation

Objective: To facilitate the creation, optimization, and maintenance of code used in the research.

Scope: Includes coding, debugging, and documentation.

Keys:

Code generation: Generating boilerplate code, scripts, or functions from natural language descriptions.

Optimization: Refactoring code for better performance or readability.

Debugging: Identifying syntax errors or logical bugs in scripts.

Automation: Writing scripts to automate file management, backups, or batch processing.

Algorithm design: Assisting in the logic and mathematical formulation of algorithms.

Documentation: Generating docstrings, comments, and README files for research software.

5. Data Management

Objective: To handle the data lifecycle from collection to reporting.

Scope: Includes cleaning, analysis, and synthetic generation. Excludes the fabrication of results (which is ethical misconduct, distinct from declared synthetic data).

Keys:

Collection: Writing scrapers or using AI agents to gather public data.

Validation: Checking data for consistency, outliers, or errors.

Cleaning: Automating the formatting, parsing, and repair of messy datasets.

Curation: Organizing and categorizing large datasets.

Analysis: Suggesting or performing statistical tests and interpreting raw outputs.

Visualization: Generating code for plots, graphs, and data dashboards.

Reporting: Summarizing data findings in textual or tabular format.

Labeling: Using LLMs to annotate or classify text/image datasets (zero-shot/few-shot labeling).

Synthesis: Generating synthetic datasets to preserve privacy or augment small samples.

Anonymization: Detecting and removing Personally Identifiable Information (PII).

Translation: Translating textual data (e.g., survey responses) into the analysis language.

6. Visuals and Multimedia

Objective: To create or enhance non-data visual elements.

Scope: Includes illustrative diagrams and image editing. Excludes scientific data plots (covered in Data Management).

Keys:

Generation: Creating conceptual images, illustrations, or diagrams from scratch.

Editing: Enhancing, cropping, or modifying existing images (e.g., removing background).

Charts/Infographics: Creating flowcharts, process diagrams, or high-level infographics.

7. Writing and Editing

Objective: To assist in the textual articulation of the research.

Scope: Includes drafting, polishing, and translation. Note: Authors remain accountable for accuracy.

Keys:

Drafting: Generating initial text for sections based on bullet points or notes.

Polishing: Correcting grammar, spelling, and punctuation.

Summarizing: Creating the abstract or plain-language summary.

Conclusions: Synthesizing the discussion into a final concluding statement.

Tone adjustment: Rewriting text to be more formal, concise, or accessible.

Translation: Translating the manuscript from the author's native language to the publication language.

References: Formatting citations and bibliography styles.

Presentation: Drafting slide decks or conference poster text.

Title generation: Brainstorming catchy and accurate titles for the work.

8. Ethics Review

Objective: To act as a check on the ethical integrity of the work.

Scope: Includes bias detection and risk assessment.

Keys:

Bias detection: Scanning text or study designs for potential cultural or gender bias.

Risk assessment: Identifying potential dual-use concerns or societal risks.

Compliance: Checking against specific ethical guidelines or checklists.

Confidentiality: Ensuring no private data is inadvertently leaked in the text.

9. Quality Assurance (Supervisor Role)

Objective: To use AI as a critical reviewer or "devil's advocate."

Scope: Includes simulated peer review and limitation checking.

Keys:

Simulated Peer Review: Using AI as a reviewer to identify unclear sections or potential critiques.

Field Trends & Consistency: Checking the manuscript against current research trends and ensuring internal consistency.

Limitations: Identifying weaknesses or limitations in the study that the authors missed.

Publication strategy: Suggesting suitable journals or conferences and helping with the submission strategy.

Publication support: Assisting in drafting cover letters, responses to reviewers, or other publication-related materials.

4

A Short LaTeX Tutorial with Examples

This Chapter aims at exemplifying how to do common stuff with LaTeX. We also show some stuff which is not that common! ;)

Please, use these examples as a starting point, but you should always consider using the *Big Oracle* (aka, Google, your best friend) to search for additional information or al-ternative ways for achieving similar results.

4.1 Document Structure

4.2 Dealing with Bibliography

Citing something online [4, 11, 6].

4.3 Inserting Tables

4.4 Importing Images

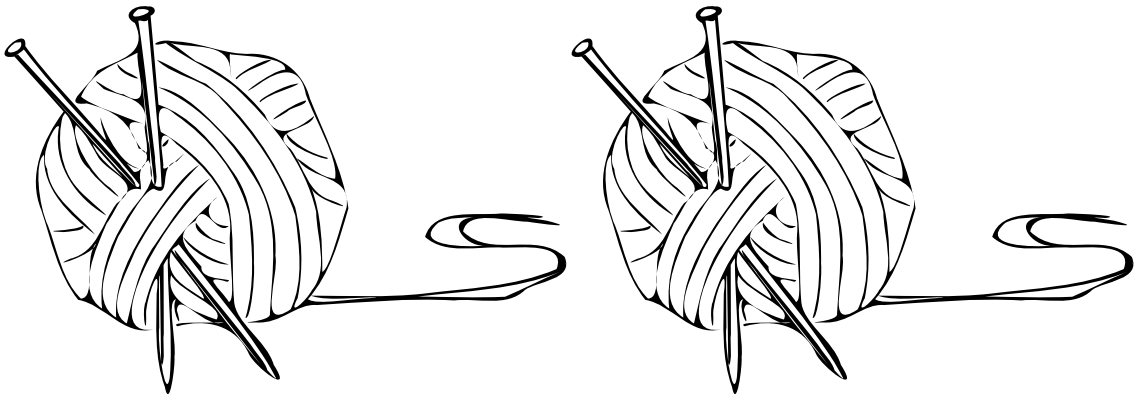
4.5 Floats, Figures and Captions

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet,

tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.



(a) One sub-figure (b) Another sub-figure

Figure 4.1: A figure with two sub-figures!

And this is a small text that references the Figure 4.1 and its Subfigures 4.1(a) and 4.1(b).

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem

non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

4.6 Text Formatting

4.7 Generating PDFs from LaTeX

4.7.1 Generating PDFs with pdflatex

You may create PDF files either by using `latex` to generate a DVI file, and then use one of the many DVI-2-PDF converters, such as `dvipdfm`.

Alternatively, you may use `pdflatex`, which will immediately generate a PDF with no intermediate DVI or PS files. In some systems, such as Apple, PDF is already the default format for LaTeX. I strongly recommend you to use this approach, unless you have a very good argument to go for `latex` + `dvipdfm`.

A typical pass for a document with figures, cross-references and a bibliography would be:

```
$ pdflatex template
$ bibtex template
$ pdflatex template
$ pdflatex template
```

You will notice that there is a new PDF file in the working directory called `template.pdf`. Simple :)

Please note that, to be sure all table of contents, cross-references and bibliographic citations are up-to-date, you must run `latex` once, then `bibtex`, and then `latex` twice.

4.7.2 Dealing with Images

You may process the same source files with both `latex` or `pdflatex`. But, if your text include images, you must be careful. `latex` and `pdflatex` accept images in different (exclusive) formats. For `latex` you may use EPS ou PS figures. For `pdflatex` you may use JPG, PNG or PDF figures. I strongly recommend you to use PDF figures in vectorial format (do not use bitmap images unless you have no other choice).

4.7.3 Dealing with Citations

To make citations, the reference key in the BibTeX file should be used. If it is a single reference [2], use a "~" to link the `\cite{...}` to the word that precedes it (...`reference~\cite{Artho04}`). If you want to make multiple citations [12, 13, 10], they should be grouped within a single `\cite{...}` and separeted by commas.

Note that the bibliography file can have as many entries as you want. Only those whose keys are referenced in the text will be included in the bibliography list.

4.7.4 Footnotes

Footnotes[1] will be numbered and shown in the bottom of the page.

4.7.5 Tables

The Table 4.1 illustrates some important concepts associated with table construction:

i) Do not use vertical lines;

ii) The caption should be above the table;

iii) Use the macros `\toprule`, `\midrule` and `\bottomrule` to make the top, inner and bottom horizontal lines, respectively.

Table 4.1: Test results summary.

Test	Anomalies	Warnings	Correct	Categories	Missed
Connection [3]	2	2	1	*C*	1
Coordinates'03 [1]	1	4	1	*2B, 1C*	0
Local Variable [1]	1	2	1	*A*	0
NASA [1]	1	1	1	—	0
Coordinates'04 [2]	1	4	1	*3C*	0
Buffer [2]	0	7	0	*2A, 1B, 2C, 2D*	0
Double-Check [2]	0	2	0	*1A, 1B*	0
StringBuffer [5]	1	0	0	—	1
Account [16]	1	1	1	—	0
Jigsaw [16]	1	2	1	*C*	0

[1] This is a simple footnote.

Over-reporting [16]	0	2	0	*1A, 1C*	0
Under-reporting [16]	1	1	1	—	0
Allocate Vector [7]	1	2	1	*C*	0
Knight Moves [3]	1	3	1	*2B*	0
Total	12	33	10	5A, 6B, 10C, 2D	2

4.7.6 Figures

The images inserted in the document must be of good quality, preferably in vector format (vector PDF) and not in *bitmap* (PNG, JPG, etc.). *bitmap* images (Figure 4.2) do not scale well and have negative effects on the quality of your document. On the other hand, *vector* images Figure 4.3 scale as much as necessary without degrading the quality of the image.

You should only use *screenshots* for your plots, charts, etc, if you absolutely have no other alternative. Instead of generating a *screenshot*, try using a virtual PDF printer and printing to a PDF file. As a general rule, you will get a vector PDF. Even if your PDF contains images, they will always be of higher or equal quality than what you would get with a *screenshot*.

To combine several figures into a single one… You can then reference the set as Figure 4.4 or the sub-figures separately as 4.4(a) and 4.4(b).

4.8 Equations

LaTeX is a powerful tool for writing in a mathematical style. It allows you to insert formulas into the text, such as this: $ax^2 + bx + c = 0$. It also allows formulas to be highlighted on a separate line and centered on the page.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

or numbered

$$e = mc^2 \tag{4.1}$$

which can later be referenced as equation 4.1

4.9 Test for listings

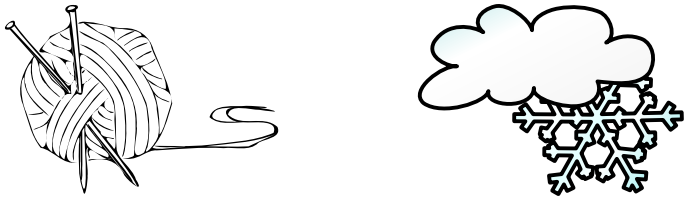
Testing the package “listings“…

Listing 4.1: cap

```
if (a==b)
  puts ("YESS!")
```

Figure 4.2: Bitmap image (JPG/PNG)

Figure 4.3: Vectorial image (PDF)



(a) Wool ball (b) Snowstorm

Figure 4.4: Example of *subbottom* usage

4.10 Test for minted

Testing the package "minted"...

```
if(a==b)
  puts("YESS!")
```

Listing 4.2: Example of a listing.

4.11 Test for algorithms

Uncomment the algorithms source below and add the following to file "`5_packages.tex`"

```
\usepackage{algorithm2e}
\RestyleAlgo{ruled}
```

and uncomment

```
\ntaddlistof{listofalgorithms}
```

in file "`6_list_of.tex`".

Bibliography

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A

The build.py Build Assistant

The file `.Build/build.py` is a Python 3 script that automates building novathesis documents from the command line without touching any configuration file by hand. It is the primary tool used by the CI pipeline and by maintainers generating sample PDFs for multiple schools at once.

A.1 Overview

The script performs four steps in sequence:

1. **Resolve the school.** Validate the `school_id` argument and auto-detect the highest available degree if `--doctype` was not supplied.
2. **Prepare a workspace.** Depending on the options, either use the current directory in-place or create a temporary directory populated with symlinks to the project tree.
3. **Patch configuration files.** If *patching mode* is active (see Section A.4), the copies of `0-Config/1_novathesis.tex` and `0-Config/4_files.tex` in the workspace are modified to reflect the requested school, document type, language, and status.
4. **Build and deliver.** Run `make` inside the workspace, then copy the resulting `template.pdf` to the output directory.

When to use `build.py` vs. `make`. For everyday writing, just run `make` (or your editor's build command). Use `build.py` when you need to build for a *different* school or document type than the one currently set in `0-Config/`, or when building from a script or CI job.

A.2 Prerequisites

- Python 3.10 or later.
- A complete TEX Live (or MiKTEX) installation with `latexmk`.

- The script must be run from the **project root** — the directory that contains `template.tex` and `0-Config/`.

```
# From the project root:
python3 .Build/build.py nova/fct
```

A.3 Quick Start

The only required argument is the *school identifier*: a `university/faculty` path that mirrors the directory structure under `NOVAthesisFiles/Schools/`.

```
# Simplest call - build for NOVA FCT in user mode
python3 .Build/build.py nova/fct

# Build a PhD thesis in English, patching configs automatically
python3 .Build/build.py nova/fct --doctype phd --lang en --docstatus final

# Build cover pages only (fast preview)
python3 .Build/build.py nova/fct --cover

# Dry run - show what would happen without building
python3 .Build/build.py nova/fct --doctype msc --dry-run
```

The separator between university and faculty may be `/`, `-`, or `.`; all are normalised to `/` internally. So `nova-fct` and `nova/fct` are identical.

A.4 Build Modes

The script operates in one of three modes, selected automatically from the options you pass.

A.4.1 User Mode

User mode is active when `--docstatus` is `keep` (the default) and neither `--cover` nor `--force-school` is given. In this mode:

- The configuration files in `0-Config/` are *not modified*.
- The build runs in the **current directory** (no temporary workspace is created).
- The result is exactly what `make` would produce.

```
python3 .Build/build.py nova/fct         # user mode (default)
python3 .Build/build.py nova/fct --progress 2  # same, with real-time output
```

A.4.2 Patching Mode

Patching mode is activated whenever any of the following options are present: `--docstatus` (with a value other than `keep`), `--cover`, or `--force-school`.

In patching mode the script:

1. Automatically creates a **temporary workspace** (unless you specify a directory with `--build-dir`).
2. Populates it with symlinks pointing back to the project tree, then copies (not symlinks) the `0-Config/` files so they can be safely modified.
3. Patches the copies to set the requested school, doctype, language, and status.
4. Builds inside the workspace; the original `0-Config/` files are never touched.

```
# Patch school, doctype, language and status - builds in a temp directory
python3 .Build/build.py nova/fct \
    --doctype phd \
    --lang en \
    --docstatus final \
    --rename-pdf \
    --output-dir build/
```

A.4.3 Cover-Only Mode

The `--cover` flag builds *only the cover pages*, skipping all chapters and bibliography. This is much faster than a full build and is useful for previewing cover layouts.

- Implies `--docstatus final` automatically.
- Comments out all `\ntaddfile` entries in `0-Config/4_files.tex` in the workspace.
- Disables the copyright page (`\ntsetup{print/copyright=false}`).
- Expected line count is reduced to ≈ 2400 for a more accurate progress bar.

```
python3 .Build/build.py nova/fct --cover
python3 .Build/build.py nova/fct --cover --lang pt --progress 2
```

A.5 Command-Line Reference

A.5.1 Positional Argument

`school_id` The school identifier in `university/faculty` format. For sub-department variants use a three-part path such as `nova/fct/cbbi`. Hyphens and dots are accepted as separators and are normalised to slashes.

Examples: `nova/fct`, `nova/fcsh`, `ulisboa/ist`, `uminho/eeng`, `nova-fct` (same as `nova/fct`).

A.5.2 Document Options

-t, --doctype {phd | msc | bsc} Document type. If omitted, the script consults `.Build/schools.conf` and selects the highest degree available for the given school (usually `phd`).

-s, --docstatus {working | provisional | final | keep} Document status written into `\ntsetup{docstatus=...}`. `keep` (the default) leaves whatever value is already in `0-Config/1_novathesis.tex` unchanged. Any other value activates patching mode.

-l, --lang *code* Two- or three-letter BCP-47 language code (default: `en`). Sets `\ntsetup{lang=...}`. Examples: `en`, `pt`, `de`, `fr`, `uk`, `gr`, `cat`.

--sdgs *n,n,...* Comma-separated list of SDG numbers to display on the cover (default: all 17). Sets `\ntsetup{print/sdgs/list=...}`.

-c, --cover Build cover pages only. Activates cover-only mode (see Section A.4.3).

-f, --force-school Force the school value to be applied even in user mode. Useful when the school in `0-Config/1_novathesis.tex` differs from the one passed on the command line.

A.5.3 Build Directory Options

-bdir / --build-dir *[path | -]* Controls where the build workspace lives:

Value	Effect
Omitted	Use current directory (user mode default)
Present, no value	Create a new `tempfile` directory
-	Reuse the directory cached in `.Build/.keep-dir.txt` from a previous run; create a new temp dir if none is cached
path	Use (or create) the specified directory

-k, --keep-bdir Do not delete the temporary workspace after a successful build. Implied when using `-bdir -` (cached directory mode).

-o, --output-dir *path* Directory where the finished PDF is copied (default: current directory). Created automatically if it does not exist.

-r, --rename-pdf Rename the output PDF from the generic `template.pdf` to `<university>-<faculty>-<do`
For example: `nova-fct-phd-en.pdf`.

A.5.4 Build Engine Options

`-p`, `--processor` *name* The `make` target used to compile LaTeX (default: `lua`). Pass `pdf` for pdfLaTeX or `xe` for XeLaTeX.

`--progress` {0 | 1 | 2} Output verbosity during compilation:

- `0` — silent (no output)
- `1` — animated progress bar (default)
- `2` — real-time `latexmk` output

`--lines` *n* Expected number of output lines for the progress-bar calculation. If omitted, the value from a previous run is read from `.Build/.keep-lines.txt`; falls back to 4400 for a full build and 2400 for cover-only mode.

A.5.5 Utility Options

`--dry-run` Print what the script *would* do — school, doctype, language, build directory, files to modify — without running the build.

`-v`, `--verbose` Enable extra debug output from the script itself.

A.6 Doctype Auto-Detection

When `--doctype` is omitted, `build.py` reads `.Build/schools.conf` to determine the highest degree available for the requested school. Each line in that file lists a school identifier followed by the degrees it supports, e.g.:

```
nova/fct     [phd, msc]  [en, pt]
uminho/ec    [phd, msc]  [en, pt]
ipl/isel     [msc]       [en, pt]
```

The script picks the *first* entry in the degree list (leftmost = highest). If the school is not listed in `schools.conf`, the fallback is `phd`. You can always override auto-detection with `--doctype`.

A.7 Examples

A.7.1 CI Pipeline Build

The GitHub Actions workflow uses:

```
python3 .Build/build.py nova/fct \
    --build-dir=AUXDIR \
    --progress=2
```

`--progress 2` streams all `latexmk` output directly to the CI log. `--build-dir AUXDIR` places all auxiliary files in a named directory so they can be archived as a workflow artifact.

A.7.2 Generating Multiple School PDFs

A shell loop can drive `build.py` to produce one PDF per school:

```
for school in nova/fct nova/fcsh ulisboa/ist uminho/eeng; do
    python3 .Build/build.py "$school"     \
        --doctype phd                     \
        --lang en                         \
        --docstatus final                 \
        --rename-pdf                      \
        --output-dir pdfs/
done
```

A.7.3 Iterative Cover Development

While working on a new school cover, iterate quickly with:

```
# First pass - creates a temp directory and caches its path
python3 .Build/build.py nova/fct --cover --build-dir -

# Subsequent passes - reuse the same temp directory
python3 .Build/build.py nova/fct --cover --build-dir -
```

Using `--build-dir -` on the first call creates a temporary directory and saves its path to `.Build/.keep-dir.txt`. All later calls with `--build-dir -` reuse that directory, skipping the symlink-creation step and making repeated builds noticeably faster.

A.7.4 Dry Run Before Committing

Before a large batch run, verify the parameters with `--dry-run`:

```
python3 .Build/build.py nova/fct \
    --doctype msc \
    --lang pt \
    --docstatus provisional \
    --dry-run
```

The script prints the resolved school, document type, language, build directory, and the configuration files it would modify, then exits without building.

A.8 Quick Reference

Option	Effect
`school_id`	Required. `university/faculty[/variant]`
`-t` / `--doctype`	`phd` / `msc` / `bsc`
`-s` / `--docstatus`	`working` / `provisional` / `final` / `keep`
`-l` / `--lang`	Language code, e.g. `en`, `pt`
`--sdgs`	SDG numbers, e.g. `1,7,13`
`-c` / `--cover`	Cover-only build (fast preview)
`-f` / `--force-school`	Apply school even in user mode
`-bdir` / `--build-dir`	Workspace: omit=current, (empty)=temp, -=cached, path
`-k` / `--keep-bdir`	Keep workspace after build
`-o` / `--output-dir`	Destination for final PDF
`-r` / `--rename-pdf`	Rename PDF to `univ-school-type-lang.pdf`
`-p` / `--processor`	LaTeX engine: `lua` (default), `pdf`, `xe`
`--progress`	`0`=silent, `1`=bar, `2`=realtime
`--lines`	Expected line count for progress bar
`--dry-run`	Preview actions without building
`-v` / `--verbose`	Extra debug output

B

NOVAthesis Macro Reference

This appendix documents the most important macros provided by the novathesis template, grouped by purpose. All signatures follow the notation used in the IntelliSense completion file `novathesis.cwl`:

- `{`⟨*arg*⟩`}` — mandatory argument
- `[`⟨*arg*⟩`]` — optional argument
- `(`⟨*arg*⟩`)` — NOVAthesis parenthesised argument (part of the command name, not a standard LaTeX delimiter)
- `*` — starred variant with modified behaviour

B.1 Template Bootstrap

`\documentclass{novathesis}`

`\documentclass[`⟨*options*⟩`]{novathesis}`
The only document class provided by novathesis. Pass engine-level or user options as a comma-separated list in the optional argument; all template-specific options are better placed in `\ntsetup` (see below).

`\ntsetup`

`\ntsetup{`⟨*key=value,...*⟩`}`

`\ntsetup[`⟨*scope*⟩`]{`⟨*key=value,...*⟩`}`
The *master configuration command*. Every template option is set here; it may be called multiple times and the last assignment wins. Place all calls in `0-Config/1_novathesis.tex`.
The most commonly used keys are listed below.

Key	Typical values	Effect
`school`	`nova/fct`, `ulisboa/ist`, …	Selects the institutional cover & strings
`doctype`	`phd`, `msc`, `bsc`, `plain`	Document type
`lang`	`en`, `pt`, `de`, …	Main document language
`media`	`screen`, `paper`	Optimises layout for on-screen or print
`docstatus`	`working`, `provisional`, `final`	Controls watermarks and certain front-matter sections

```
\ntsetup{school=nova/fct, doctype=phd, lang=en, media=screen}
\ntsetup{docstatus=working}
```

B.2 Document Identity

`\nttitle`

`\nttitle(`⟨*type*⟩`,`⟨*lang*⟩`){`⟨*title*⟩`}`

Sets one of the document's titles. Different *type* variants are used in different locations (cover page, spine, abstract header, etc.) and can be provided in multiple languages.

type	Where used
`main`	Cover page, headers, PDF metadata
`sub`	Subtitle
`cover`	Overrides the cover title only
`abstract`	Abstract page header
`spine`	Spine label (usually a short form)

lang is any supported language code: `en`, `pt`, `de`, `es`, `fr`, `it`, `nl`, `pl`, `cz`, `dk`, `sk`, `cat`, `gr`, `uk`, `zhs`, `zht`.

```
\nttitle(main,en){Thesis Title in English}
\nttitle(main,pt){Título da Tese em Português}
\nttitle(spine,en){Short Title}
```

`\ntauthorname` `\ntauthordegree`

`\ntauthorname(`⟨*gender*⟩`){`⟨*full name*⟩`}{`⟨*short name*⟩`}`

`\ntauthordegree(`⟨*lang*⟩`){`⟨*degree*⟩`}`

`\ntauthorname` registers the author's full and short names. The *gender* argument (`m`

or `f`) controls grammatical agreement in Portuguese strings.
`\ntauthordegree` sets the author's pre-existing academic degree for display on the cover.

```
\ntauthorname(m){João Maria Lourenço}{João Lourenço}
\ntauthordegree(en){Bachelor in Computer Science}
\ntauthordegree(pt){Licenciado em Engenharia Informática}
```

`\ntdate`

`\ntdate(`⟨*type*⟩`){`⟨*YYYY-MM-DD*⟩`}`

`\SetDateISO(`⟨*type*⟩`){`⟨*YYYY-MM-DD*⟩`}`

`\PrintDateISO{`⟨*YYYY-MM-DD*⟩`}`

`\todayiso`

`\ntdate` registers a document date. *type* is either `submission` (date sent to the jury) or `exam` (public defence date).
`\SetDateISO` / **`\PrintDateISO`** are lower-level helpers for ISO 8601 date handling elsewhere in the document. **`\todayiso`** expands to today's date in ISO format.
The date string may be `YYYY-MM-DD`, `YYYY-MM`, or just `YYYY`.

```
\ntdate(submission){2025-09-30}
\ntdate(exam){2025-11-14}
```

B.3 Institutional Metadata

`\ntdepartment` **`\ntdegreename`** **`\ntspecialization`**

`\ntdepartment(`⟨*lang*⟩`){`⟨*name*⟩`}`

`\ntdepartment*(`⟨*lang*⟩`){`⟨*name*⟩`}`

`\ntdegreename(`⟨*lang*⟩`){`⟨*name*⟩`}`

`\ntdegreename*(`⟨*lang*⟩`){`⟨*name*⟩`}`

`\ntspecialization(`⟨*lang*⟩`){`⟨*name*⟩`}`

`\ntspecialization*(`⟨*lang*⟩`){`⟨*name*⟩`}`

Override the school-default department name, degree name, or specialisation string. Provide values for every language used in the document (at minimum the document language and Portuguese for national universities). The starred form sets an abbreviated variant used on the spine and narrow covers.

```
\ntdepartment(en){Department of Computer Science}
\ntdepartment(pt){Departamento de Informática}
```

```
\ntdegreename(en){Doctor of Philosophy in Computer Science}
\ntdegreename(pt){Doutor em Engenharia Informática}
```

\ntsponsors

\ntsponsors(⟨*order*⟩,⟨*type*⟩,⟨*lang*⟩){⟨*content*⟩}
Registers a funding sponsor for display on the cover. *order* is a positive integer that controls display order. *type* is either `text` (a string) or `logo` (an `\includegraphics` call or similar).

```
\ntsponsors(1,logo,en){\includegraphics[height=1cm]{fct-logo}}
\ntsponsors(2,text,en){Supported by FCT grant XXXXXX/XX/XXXXX}
```

B.4 People — Advisers and Committee

\ntaddperson

\ntaddperson{⟨*class*⟩}(⟨*role*⟩,⟨*gender*⟩){⟨*name, position, institution*⟩}
Registers a person (adviser or jury member). All fields are typeset verbatim on the cover and in the jury table.

class	role	Meaning
adviser	a	Principal adviser
	c / co	Co-adviser
	t	Tutor
committee	c	Chair
	r	Rapporteur
	a	Adviser
	co	Co-adviser
	m	Member
	g	Guest

The *gender* argument (`m` / `f`) governs grammatical agreement in Portuguese texts.

```
\ntaddperson{adviser}(a,m){%
  João Maria Lourenço, Full Professor,
  NOVA School of Science and Technology, UNL}
\ntaddperson{committee}(c,m){%
  Maria Fernanda Távora, Full Professor,
```

```
Instituto Superior Técnico, ULisboa}
```

\ntprintpersons

\ntprintpersons{⟨*class*⟩}{⟨*roles*⟩}

\GetPersonsAsList{⟨*class*⟩}{⟨*roles*⟩}[⟨*separator*⟩]

\ntprintpersons typesets a formatted table of persons filtered by class and role(s). Use it inside cover or front-matter files to display the jury or adviser list.

\GetPersonsAsList returns the same persons as a flat list with an optional custom separator.

```
\ntprintpersons{adviser}{a,co}
\ntprintpersons{committee}{c,r,m}
```

B.5 File Registration — \ntaddfile

\ntaddfile

\ntaddfile{⟨*class*⟩}{⟨*filename*⟩}

\ntaddfile{⟨*class*⟩}[⟨*key*⟩]{⟨*filename*⟩}

The single command used to register *every* user-provided file with the template. Files are rendered later by the corresponding \ntprint* call in template.tex. The *filename* must be given without extension; the .tex suffix is appended automatically. Multiple calls accumulate; files are printed in registration order.

class	key	Notes
bib	—	BibTeX/Biber .bib database
abstract	language code (en, pt, …)	One per language; key selects the language
glossaries	glossary | acronym | symbols	Glossary-type files loaded by glossaries-extra
chapter	optional doctype list	Main-matter chapters; key restricts to listed doctypes
appendix	optional doctype list	Back-matter appendices
annex	optional doctype list	Back-matter annexes
cover	1 | 2 | *N* | spine	Overrides the built-in cover layout for slot *N*
dedication	—	
acknowledgements	—	
quote	—	
statement	language code	Signed statement; key selects the language

```
\ntaddfile{bib}{bibliography}
\ntaddfile{abstract}[en]{abstract-en}
\ntaddfile{abstract}[pt]{abstract-pt}
\ntaddfile{glossaries}[acronym]{acronyms}
\ntaddfile{chapter}{chapter-introduction}
\ntaddfile{chapter}[phd,msc]{chapter-advanced}% only in PhD and MSc
\ntaddfile{appendix}{app-build-py}
\ntaddfile{annex}{annex-data}
```

B.6 Content Rendering

\ntprint \ntprintlist

\ntprint{⟨*key*⟩}

\ntprintlist{⟨*key*⟩}

High-level rendering commands used exclusively in template.tex. **\ntprint** outputs a single named section; **\ntprintlist** outputs all items registered under the given key (e.g., all chapters).

These abstract away the complex logic of conditionally printing front-matter elements, list-of-figures, etc. depending on the current docstatus and doctype.

Common key values include: frontcover, frontmatter, mainmatter, backmatter,

```
backcover.
```

\ntaddlistof

\ntaddlistof{⟨*listof*⟩}

\ntaddlistof[⟨*matter*⟩]{⟨*listof*⟩}

Registers a "List of …" to be printed by \ntprint{frontmatter} or \ntprint{backmatter}. The optional ⟨*matter*⟩ argument is either frontmatter (default) or backmatter, and controls where the list is typeset. ⟨*listof*⟩ is the internal identifier of the list (e.g. figures, tables, listings).

```
% Print list of figures in front matter (default)
\ntaddlistof{figures}
% Print list of listings in back matter
\ntaddlistof[backmatter]{listings}
```

B.7 Cover Customisation

\ntaddtocover

\ntaddtocover{⟨*cover-ids*⟩}{⟨*content*⟩}

\ntaddtocover[⟨*scope*⟩]{⟨*cover-ids*⟩}{⟨*content*⟩}

Appends *content* to one or more cover slots identified by *cover-ids* (a comma-separated list of slot identifiers such as 1, 2, spine, N). Use the optional *scope* to restrict the addition to a particular school or document type context.

```
\ntaddtocover{1,2}{\textbf{CONFIDENTIAL} - not for distribution}
```

\ntAddToCoverTikz

\ntAddToCoverTikz(⟨*layer*⟩){⟨*cover-ids*⟩}{⟨*tikz-code*⟩}

\ntAddToCoverTikz<⟨*label*⟩>[⟨*tikz-options*⟩](⟨*layer*⟩)[⟨*more-opts*⟩]{⟨*cover-ids*⟩}{⟨*tikz-code*⟩}

Injects raw TikZ drawing code onto the cover canvas at the specified *layer* (e.g. bg, fg, text). Coordinates use the full cover dimensions. Use this for advanced cover layouts in school .ldf files.

```
\ntAddToCoverTikz(bg){1}{%
  \fill[blue!20] (0,0) rectangle (\paperwidth,3cm);
}
```

B.8 Document Conditionals

\ifphddoc, \ifmscdoc, …

\ifdocclass{⟨*class*⟩}{⟨*true*⟩}{⟨*false*⟩}

\ifphddoc{⟨*true*⟩}{⟨*false*⟩}

\ifmscdoc{⟨*true*⟩}{⟨*false*⟩}

\ifbscdoc{⟨*true*⟩}{⟨*false*⟩}

\ifplaindoc{⟨*true*⟩}{⟨*false*⟩}

\ifdocstatus{⟨*status*⟩}{⟨*true*⟩}{⟨*false*⟩}

\ifdocfinal{⟨*true*⟩}{⟨*false*⟩}

\ifdocprovisional{⟨*true*⟩}{⟨*false*⟩}

\ifdocworking{⟨*true*⟩}{⟨*false*⟩}

\ifcoverdefined{⟨*cover-id*⟩}{⟨*true*⟩}{⟨*false*⟩}

\ifxeorlua{⟨*xe/lua code*⟩}{⟨*pdflatex code*⟩}

A family of Boolean predicates for conditional content. All follow the same two-argument form: *true code* and *false code*. The *false* argument may be omitted (empty by default).

```
% Acknowledgements only in the final version
\ifdocfinal{%
  \ntaddfile{acknowledgements}{acknowledgements}%
}

% PhD-specific chapter
\ifphddoc{%
  \ntaddfile{chapter}[phd]{chapter-related-work}%
}

% Engine-dependent font choice
\ifxeorlua{%
  \setmainfont{TeX Gyre Pagella}%
}{%
  \usepackage{newpxtext}%
}
```

B.9 Hook System

\NTAddToHook \NTDeclareHook \NTRunHook

\NTDeclareHook{⟨*hook-name*⟩}

\NTAddToHook{⟨*hook-name*⟩}{⟨*code*⟩}

\NTRunHook{⟨*hook-name*⟩}

novathesis exposes a lightweight hook system used primarily in school `.ldf` files to inject code at well-defined points (cover rendering, font loading, language setup, etc.) without modifying the class source.

Hooks are declared once with **\NTDeclareHook**, populated with **\NTAddToHook** (can be called multiple times — code accumulates), and executed at the right moment with **\NTRunHook**.

```
% In a school .ldf file:
\NTAddToHook{cover/before}{\MySchoolSetupCode}
\NTAddToHook{fonts/load}{\setmainfont{TeX Gyre Pagella}}
```

B.10 Miscellaneous Utilities

\shiftbookmarks

\shiftbookmarks{⟨*content*⟩}

\shiftbookmarks[⟨*levels*⟩]{⟨*content*⟩}

Shifts all PDF bookmark levels inside *content* by the given number of *levels* (default: 1). Used at the start of back matter to nest appendix and annex bookmarks one level below the "Back Matter" root bookmark.

```
\shiftbookmarks[1]{\ntprintbibsingle \ntprintappendices \ntprintannexes}
```

\ntselectlang

\ntselectlang{⟨*lang-code*⟩}

Switches the active Babel / Polyglossia language mid-document and updates all novathesis localised strings accordingly. Used internally when rendering multi-language abstracts; available for advanced users who need language changes inside chapters.

```
\ntselectlang{pt}
\noindent\textbf{Resumo} - \lipsum[1]
\ntselectlang{en}
```

\debuggrid

\debuggrid

\debuggrid[⟨*color*⟩]
Overlays a grid on the current page to aid layout debugging. Default colour is `red!30`. Wrap in a local group or remove before final submission.

```
\debuggrid[blue!20]   % draw a blue grid on the current page
```

\novathesis

\novathesis
Typesets the template name novathesis in the canonical stylised form. Use it whenever referring to the template inside your document.

```
This document was prepared using the \novathesis{} template.
```

B.11 Summary Table

The table below lists all documented macros with a short description and the configuration file where they are typically used.

Macro	Purpose	Typical location
\documentclass {novathesis}	Load the template class	template.tex
\ntsetup	Master document configuration	0-Config/1_novathesis.tex
\nttitle	Set title (type + language)	0-Config/3_cover.tex
\ntauthorname	Author full and short name	0-Config/3_cover.tex
\ntauthordegree	Author's pre-existing degree	0-Config/3_cover.tex
\ntdate	Submission / exam date	0-Config/3_cover.tex
\ntdepartment	Override department name	0-Config/3_cover.tex
\ntdegreename	Override degree name	0-Config/3_cover.tex
\ntspecialization	Override specialisation name	0-Config/3_cover.tex
\ntsponsors	Funding sponsor (text or logo)	0-Config/3_cover.tex
\ntaddperson	Register adviser or jury member	0-Config/3_cover.tex
\ntprintpersons	Typeset adviser/jury table	Cover .tex files
\ntaddfile	Register any user file with the template	0-Config/4_files.tex
\ntprint	Print one named document section	*Internal*

(continued on next page)

Macro	Purpose	Typical location
`\ntprintlist`	Print all files in a registered list	`Internal`
`\ntaddlistof`	Register a "List of …"; optional arg `frontmatter` (default) or `backmatter` sets where it prints	`0-Config/4_files.tex`
`\ntaddtocover`	Append content to cover slot(s)	`School .clo files`
`\ntAddToCoverTikz`	Inject Ti*k*Z code onto cover canvas	`School .clo files`
`\ifphddoc` / `\ifmscdoc`	Conditional on document type	`0-Config/4_files.tex, chapters`
`\ifdocfinal`	Conditional on document status	`0-Config/4_files.tex, chapters`
`\ifxeorlua`	Conditional on LaTeX engine	`0-Config/5_packages.tex`
`\NTDeclareHook`	Declare an extension hook	`School .clo files`
`\NTAddToHook`	Add code to an extension hook	`School .clo files`
`\NTRunHook`	Execute an extension hook	`Class internals`
`\shiftbookmarks`	Shift PDF bookmark levels	`nt-render-matter.sty`
`\ntselectlang`	Switch active language mid-document	`Abstract files, chapters`
`\debuggrid`	Overlay layout debug grid	`During development`
`\novathesis`	Template name in canonical style	`Anywhere in text`

I

Lorem Ipsum Annex

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

I.1 A section A

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

I.2 A section B

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus.

Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

I.2.1 A section B.1

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

I.2.2 A section B.2

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

2000 A Very Long and Impressive Thesis Title with a Forced Line Break John Doe